

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment**

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|                       |                                                                                                                  |                      |                                                 |
|-----------------------|------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------|
| <b>Course Code:</b>   | CSA10                                                                                                            | <b>Course Title:</b> | Software Engineering                            |
| <b>CO Assessed:</b>   | CO4                                                                                                              | <b>Assessment:</b>   | Assessment Tool 1 – Test Case Design Assignment |
| <b>Weightage:</b>     | 20%                                                                                                              | <b>Total Marks:</b>  | 25                                              |
| <b>CO4 Statement:</b> | Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics |                      |                                                 |
| <b>Student Name:</b>  | Mukesh Raj L                                                                                                     | <b>Reg. No.:</b>     | 192571036                                       |
| <b>Date:</b>          |                                                                                                                  | <b>Duration:</b>     | 60 minutes                                      |

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**Annexure A – Test Case Design Assignment**

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**Instructions to Students:**

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

**Deliverable:** Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

## Rubrics for Calculation

| Evaluation Criteria                                        | Excellent (4 Marks)                                                                                                                   | Good (3 Marks)                                     | Satisfactory (2 Marks)                                      | Needs Improvement (1 Mark)                                   |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------|
| <b>Requirement &amp; Test Scenario Identification(15%)</b> | Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios                               | Identifies most requirements and scenarios         | Identifies basic requirements and scenarios                 | Requirements/scenarios are incomplete or unclear             |
| <b>Test Case Design &amp; Completeness(25%)</b>            | Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions                                               | Covers most important conditions with minor gaps   | Covers basic conditions but misses several important cases  | Test cases are very limited or incomplete                    |
| <b>Test Data &amp; Expected Results(20%)</b>               | Appropriate test data with precise, measurable expected results for every test case                                                   | Mostly appropriate test data and expected results  | Some test data/results are unclear or incomplete            | Test data and expected results are largely missing/incorrect |
| <b>Application of Test Design Techniques(15%)</b>          | Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition | Correctly applies at least two relevant techniques | Applies one technique with limited justification            | Techniques are not applied or incorrectly applied            |
| <b>Traceability &amp; Test Coverage(15%)</b>               | Excellent mapping between requirements, scenarios, and test cases with strong coverage                                                | Good traceability and coverage with minor gaps     | Partial traceability and moderate coverage                  | Poor/no traceability and inadequate coverage                 |
| <b>Documentation &amp; Presentation(10%)</b>               | Well-structured, clear, professional, consistent, and easy to understand                                                              | Well-organized with minor presentation issues      | Adequately organized but has some clarity/formatting issues | Poorly organized, unclear, or incomplete documentation       |

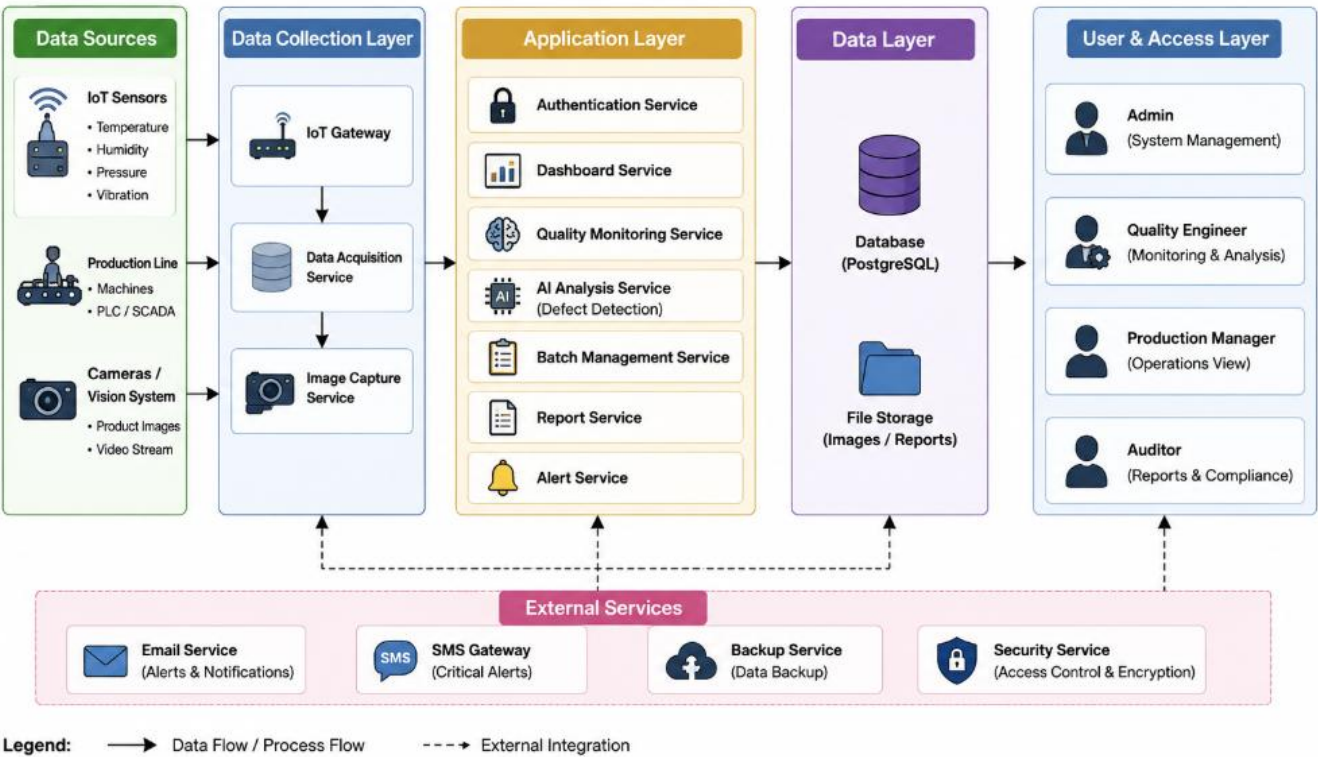
| Criteria                                                    | Mark |
|-------------------------------------------------------------|------|
| <b>Requirement &amp; Test Scenario Identification (15%)</b> | /5   |
| <b>Test Case Design &amp; Completeness (25%)</b>            | /4   |
| <b>Test Data &amp; Expected Results (20%)</b>               | /4   |
| <b>Application of Test Design Techniques (15%)</b>          | /4   |
| <b>Traceability &amp; Test Coverage (15%)</b>               | /4   |
| <b>Documentation &amp; Presentation (10%)</b>               | /4   |
| <b>TOTAL</b>                                                | /25  |

# 1.INTRODUCTION

The AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform is designed to improve the monitoring and quality assurance process in pharmaceutical manufacturing. The system integrates production monitoring, AI-based defect detection, quality prediction, batch traceability, alerts, dashboard monitoring, and reporting.

The purpose of this test case design document is to identify the requirements and test scenarios of the proposed system and prepare comprehensive test cases to verify its correctness, functionality, reliability, and usability.

According to the assignment, the test cases should include functional and non-functional requirements, test scenarios, normal, boundary, invalid, and exceptional conditions, along with Test Case ID, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status. Appropriate techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition Testing should also be applied.



## 2. FUNCTIONAL REQUIREMENTS

The following functional requirements are identified for the system.

| Requirement ID | Functional Requirement                                                        |
|----------------|-------------------------------------------------------------------------------|
| FR01           | The system shall allow authorized users to log in.                            |
| FR02           | The system shall validate user credentials.                                   |
| FR03           | The system shall display the pharmaceutical quality dashboard.                |
| FR04           | The system shall collect and display production parameters.                   |
| FR05           | The system shall detect abnormal production conditions.                       |
| FR06           | The system shall perform AI-based product defect detection.                   |
| FR07           | The system shall predict possible quality deviations.                         |
| FR08           | The system shall generate alerts for abnormal conditions or detected defects. |
| FR09           | The system shall store and retrieve batch information.                        |
| FR10           | The system shall support batch traceability.                                  |
| FR11           | The system shall generate quality/compliance reports.                         |
| FR12           | The system shall allow authorized users to log out safely.                    |

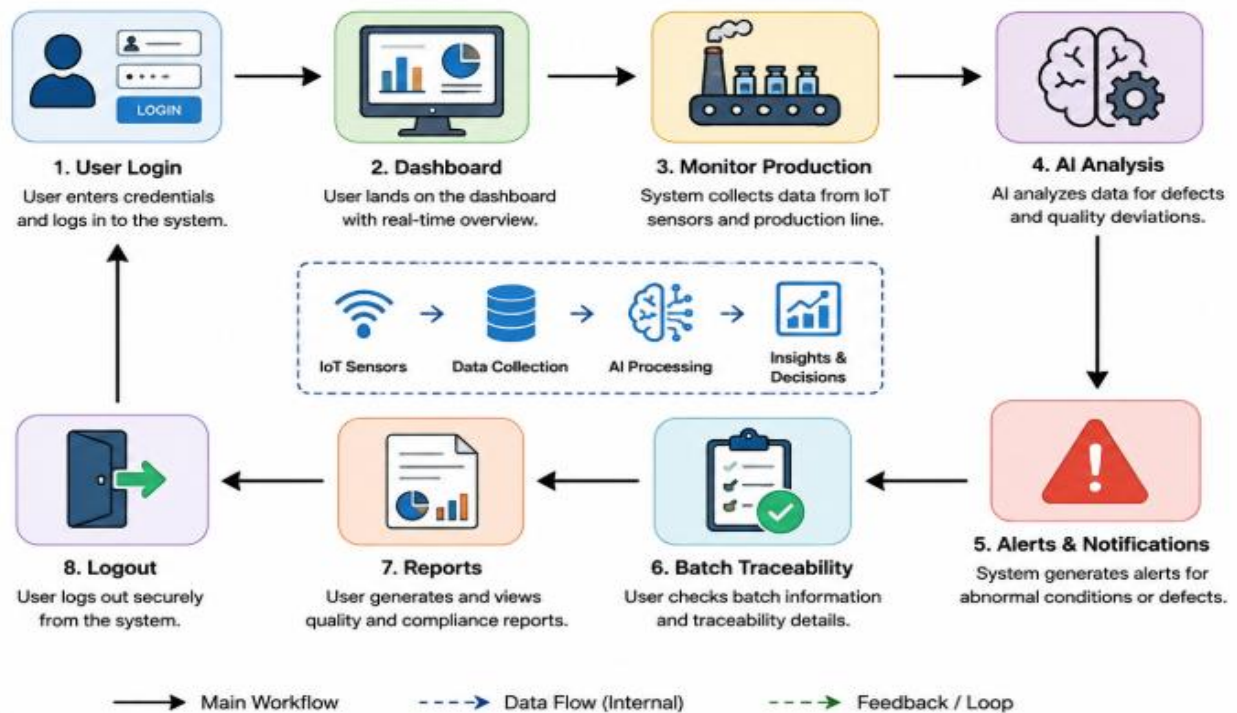
## 3. NON-FUNCTIONAL REQUIREMENTS

| Requirement ID | Non-Functional Requirement                                                                          |
|----------------|-----------------------------------------------------------------------------------------------------|
| NFR01          | The system should provide a user-friendly interface.                                                |
| NFR02          | The system should respond to valid user requests within an acceptable time.                         |
| NFR03          | The system should handle invalid input without crashing.                                            |
| NFR04          | Unauthorized users should not access protected system functions.                                    |
| NFR05          | Stored data should remain available and reliable.                                                   |
| NFR06          | The system should provide meaningful error messages.                                                |
| NFR07          | The system should maintain consistency of batch and quality information.                            |
| NFR08          | The system should continue operating correctly when valid data is provided.                         |
| NFR09          | The system should support maintainability through modular components.                               |
| NFR10          | The system should safely handle exceptional conditions such as unavailable data or failed requests. |

## 4. TEST SCENARIOS

The following major test scenarios are identified.

| Scenario ID | Test Scenario                | Related Requirement |
|-------------|------------------------------|---------------------|
| TS01        | User Login                   | FR01, FR02          |
| TS02        | Invalid Login Handling       | FR02, NFR03         |
| TS03        | Dashboard Access             | FR03                |
| TS04        | Production Data Monitoring   | FR04                |
| TS05        | Abnormal Parameter Detection | FR05                |
| TS06        | AI-Based Defect Detection    | FR06                |
| TS07        | Quality Deviation Prediction | FR07                |
| TS08        | Alert Generation             | FR08                |
| TS09        | Batch Information Storage    | FR09                |
| TS10        | Batch Traceability           | FR10                |
| TS11        | Report Generation            | FR11                |
| TS12        | User Logout                  | FR12                |
| TS13        | Boundary Value Validation    | NFR03, NFR06        |
| TS14        | Invalid Input Handling       | NFR03, NFR06        |
| TS15        | System Reliability           | NFR05, NFR08        |
| TS16        | Unauthorized Access          | NFR04               |
| TS17        | Exception Handling           | NFR10               |



## 5. DETAILED TEST CASES

5.1 User Authentication Test Cases



**AI-Based Smart Pharmaceutical Manufacturing**  
Quality Assurance Platform

Ensuring Quality, Safety and Compliance

## Welcome Back!

Sign in to your account

**Username**

 Enter your username

**Password**

 Enter your password 

☐ Remember me

[Forgot Password?](#)

→ Login

OR

 Sign in with Google

 Unauthorized access is strictly prohibited.  
All activities are monitored and logged.

TC01 – Valid User Login

| Field           | Details                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------|
| Test Case ID    | TC01                                                                                          |
| Test Scenario   | Valid User Login                                                                              |
| Test Steps      | 1. Open the login page. 2. Enter a valid username. 3. Enter a valid password. 4. Click Login. |
| Test Data       | Valid username and valid password                                                             |
| Expected Result | User should be successfully authenticated and redirected to the dashboard.                    |
| Actual Result   | To be filled after testing                                                                    |
| Status          | To be filled as Pass/Fail                                                                     |

TC02 – Invalid Password

| Field         | Details                                                                                   |
|---------------|-------------------------------------------------------------------------------------------|
| Test Case ID  | TC02                                                                                      |
| Test Scenario | Login with Invalid Password                                                               |
| Test Steps    | 1. Open login page. 2. Enter valid username. 3. Enter incorrect password. 4. Click Login. |
| Test Data     | Valid username + invalid password                                                         |

| Field           | Details                                                              |
|-----------------|----------------------------------------------------------------------|
| Expected Result | System should reject login and display an appropriate error message. |
| Actual Result   | To be filled after testing                                           |
| Status          | To be filled as Pass/Fail                                            |

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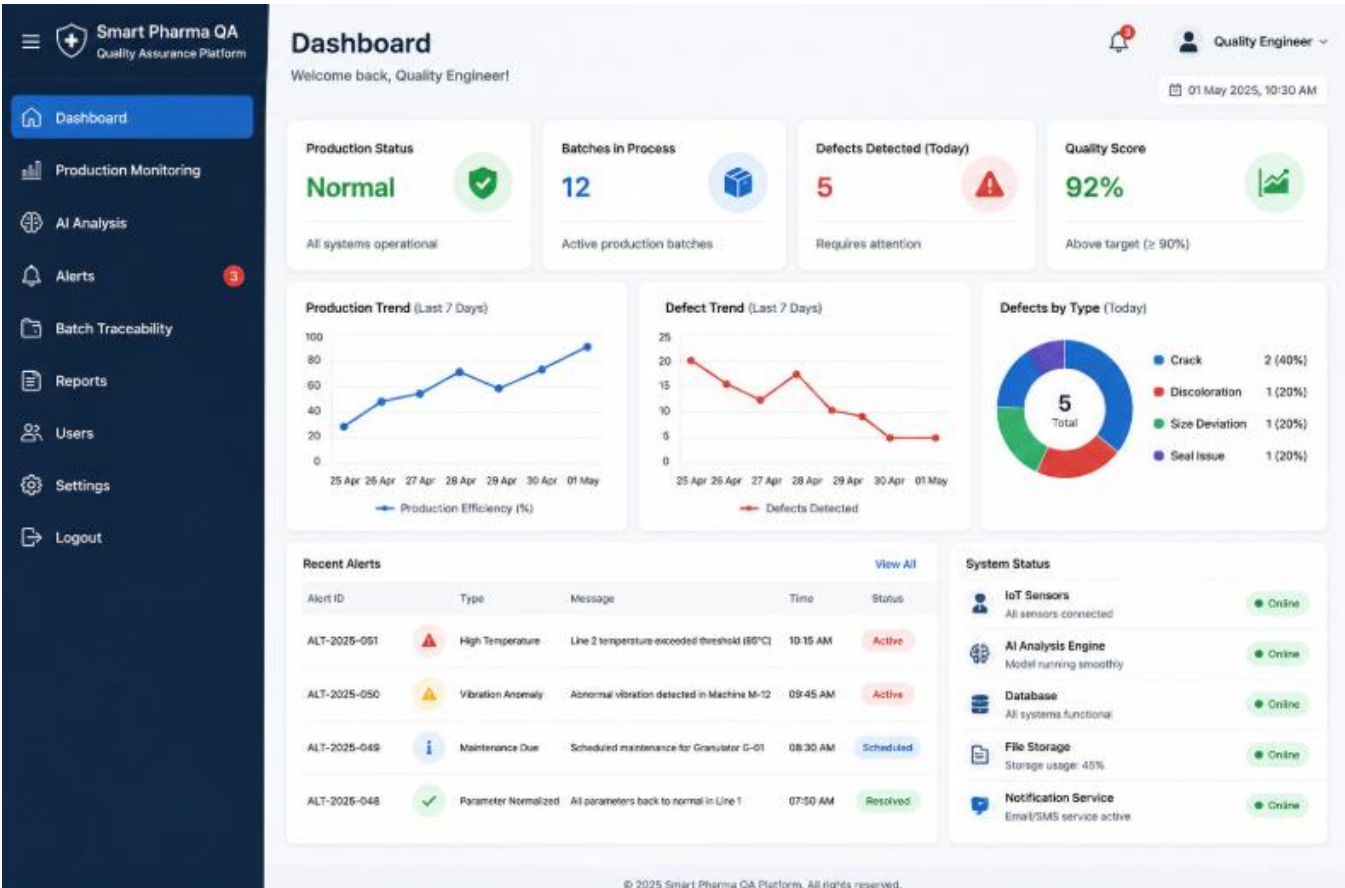
### TC03 – Invalid Username

| Field           | Details                                                             |
|-----------------|---------------------------------------------------------------------|
| Test Case ID    | TC03                                                                |
| Test Scenario   | Login with Invalid Username                                         |
| Test Steps      | 1. Enter invalid username. 2. Enter password. 3. Click Login.       |
| Test Data       | Invalid username                                                    |
| Expected Result | System should deny access and display an appropriate error message. |
| Actual Result   | To be filled after testing                                          |
| Status          | To be filled as Pass/Fail                                           |

### TC04 – Empty Login Fields

| Field           | Details                                                               |
|-----------------|-----------------------------------------------------------------------|
| Test Case ID    | TC04                                                                  |
| Test Scenario   | Login with Empty Fields                                               |
| Test Steps      | 1. Leave username empty. 2. Leave password empty. 3. Click Login.     |
| Test Data       | Empty username and password                                           |
| Expected Result | System should display validation messages and should not allow login. |
| Actual Result   | To be filled after testing                                            |
| Status          | To be filled as Pass/Fail                                             |

## 5.2 Dashboard Test Cases



### TC05 – Dashboard Loading

| Field           | Details                                                                      |
|-----------------|------------------------------------------------------------------------------|
| Test Case ID    | TC05                                                                         |
| Test Scenario   | Open Quality Dashboard                                                       |
| Test Steps      | 1. Log in with valid credentials. 2. Navigate to dashboard.                  |
| Test Data       | Valid authenticated user                                                     |
| Expected Result | Dashboard should load successfully and display available system information. |
| Actual Result   | To be filled after testing                                                   |
| Status          | To be filled as Pass/Fail                                                    |

### TC06 – Unauthorized Dashboard Access

| Field         | Details                                                                                     |
|---------------|---------------------------------------------------------------------------------------------|
| Test Case ID  | TC06                                                                                        |
| Test Scenario | Access Dashboard Without Login                                                              |
| Test Steps    | 1. Log out or open a new session. 2. Attempt to access a protected dashboard page directly. |

| Field           | Details                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| Test Data       | Unauthenticated user                                                                                         |
| Expected Result | System should deny access and redirect the user to the login page or display an unauthorized access message. |
| Actual Result   | To be filled after testing                                                                                   |
| Status          | To be filled as Pass/Fail                                                                                    |

## 5.3 Production Monitoring Test Cases

### TC07 – Valid Production Data

| Field           | Details                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------|
| Test Case ID    | TC07                                                                                       |
| Test Scenario   | Display Valid Production Parameters                                                        |
| Test Steps      | 1. Provide valid production data. 2. Submit or receive data. 3. Open monitoring dashboard. |
| Test Data       | Valid production parameter values                                                          |
| Expected Result | System should accept and correctly display the production information.                     |
| Actual Result   | To be filled after testing                                                                 |
| Status          | To be filled as Pass/Fail                                                                  |

### TC08 – Missing Production Data

| Field           | Details                                                                                |
|-----------------|----------------------------------------------------------------------------------------|
| Test Case ID    | TC08                                                                                   |
| Test Scenario   | Handle Missing Production Data                                                         |
| Test Steps      | 1. Send incomplete or missing production data. 2. Attempt processing.                  |
| Test Data       | Missing required parameter                                                             |
| Expected Result | System should not crash and should display an appropriate validation or error message. |
| Actual Result   | To be filled after testing                                                             |
| Status          | To be filled as Pass/Fail                                                              |

TC09 – Abnormal Production Condition

| Field           | Details                                                                             |
|-----------------|-------------------------------------------------------------------------------------|
| Test Case ID    | TC09                                                                                |
| Test Scenario   | Detect Abnormal Condition                                                           |
| Test Steps      | 1. Provide production data representing an abnormal condition. 2. Process the data. |
| Test Data       | Parameter value outside the configured acceptable range                             |
| Expected Result | System should identify the abnormal condition and generate an appropriate alert.    |
| Actual Result   | To be filled after testing                                                          |
| Status          | To be filled as Pass/Fail                                                           |

5.4 AI-Based Defect Detection Test Cases

Smart Pharma QA  
Quality Assurance Platform

Dashboard

Production Monitoring

AI Analysis

Alerts

Batch Traceability

Reports

Users

Settings

Logout

AI Defect Detection

Analyzed Image



DETECT: Crack

Detected Regions



Crack



Discoloration



No Defect



No Defect

Recommendations



- Immediate inspection of items in the affected batch.
- Check Line 2 conveyor alignment and vibration.
- Review camera and filter camera parameters.

Analysis Result



DEFECT DETECTED

Defect Type : Crack

Confidence : 94.6%

All Detected Defects

| Defect Type    | Confidence | Severity |
|----------------|------------|----------|
| Crack          | 94.6%      | High     |
| Discoloration  | 87.2%      | Medium   |
| Size Deviation | 76.5%      | Low      |
| Seal Issue     | 72.1%      | Low      |

Image Information

Image Name

: vial\_batch\_20250501\_1030.jpg

Captured On

: 01 May 2025, 10:30 AM

Batch ID

: B245

Line

: Line 2

Model Used

: PharmaDefectNet v2.1

Status

: Completed

TC10 – Valid Product Image Analysis

| Field         | Details                     |
|---------------|-----------------------------|
| Test Case ID  | TC10                        |
| Test Scenario | Analyze Valid Product Image |

| Field           | Details                                                                            |
|-----------------|------------------------------------------------------------------------------------|
| Test Steps      | 1. Upload a supported valid product image. 2. Start AI analysis.                   |
| Test Data       | Valid image file                                                                   |
| Expected Result | System should accept the image and return a defect detection or inspection result. |
| Actual Result   | To be filled after testing                                                         |
| Status          | To be filled as Pass/Fail                                                          |

#### TC11 – Unsupported Image File

| Field           | Details                                                                  |
|-----------------|--------------------------------------------------------------------------|
| Test Case ID    | TC11                                                                     |
| Test Scenario   | Upload Unsupported File                                                  |
| Test Steps      | 1. Select an unsupported file format. 2. Upload the file for inspection. |
| Test Data       | Unsupported file format                                                  |
| Expected Result | System should reject the file and display a suitable validation message. |
| Actual Result   | To be filled after testing                                               |
| Status          | To be filled as Pass/Fail                                                |

#### TC12 – Corrupted Image File

| Field           | Details                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------|
| Test Case ID    | TC12                                                                                       |
| Test Scenario   | Handle Corrupted Image                                                                     |
| Test Steps      | 1. Upload a corrupted or unreadable image file. 2. Start analysis.                         |
| Test Data       | Corrupted image                                                                            |
| Expected Result | System should safely handle the error without crashing and display an appropriate message. |
| Actual Result   | To be filled after testing                                                                 |
| Status          | To be filled as Pass/Fail                                                                  |

## 5.5 Quality Prediction Test Cases

#### TC13 – Valid Quality Prediction

| Field           | Details                                                                |
|-----------------|------------------------------------------------------------------------|
| Test Case ID    | TC13                                                                   |
| Test Scenario   | Predict Quality Deviation                                              |
| Test Steps      | 1. Provide valid input data. 2. Submit data to prediction module.      |
| Test Data       | Valid production and quality parameters                                |
| Expected Result | System should process the data and return a quality prediction result. |
| Actual Result   | To be filled after testing                                             |
| Status          | To be filled as Pass/Fail                                              |

## TC14 – Invalid Prediction Input

| Field           | Details                                                                     |
|-----------------|-----------------------------------------------------------------------------|
| Test Case ID    | TC14                                                                        |
| Test Scenario   | Invalid Input to Prediction Module                                          |
| Test Steps      | 1. Enter invalid or incomplete values. 2. Start prediction.                 |
| Test Data       | Invalid or incomplete data                                                  |
| Expected Result | System should reject invalid data and provide an appropriate error message. |
| Actual Result   | To be filled after testing                                                  |
| Status          | To be filled as Pass/Fail                                                   |

## 5.6 Alert Generation Test Cases

Smart Pharma QA

Quality Assurance Platform

Dashboard

Production Monitoring

AI Analysis

Alerts 3

Batch Traceability

Reports

Users

Settings

Logout

Alerts & Notifications

View and manage system alerts in real-time.

All Status

01 May 2025 - 01 May 2025

Search by Alert ID / Type / Message...

Critical Alerts

2

Requires immediate action

Warning Alerts

4

Requires attention

Info Alerts

7

For information only

Resolved Today

5

Alerts resolved

Recent Alerts

View All Alerts

| Alert ID     | Type     | Message                                       | Source                      | Time                    | Status    |
|--------------|----------|-----------------------------------------------|-----------------------------|-------------------------|-----------|
| ALT-2025-051 | Critical | Line 2 temperature exceeded threshold (85°C)  | Temperature Sensor (Line 2) | 10:15 AM<br>01 May 2025 | Active    |
| ALT-2025-050 | Warning  | Abnormal vibration detected in Machine M-12   | Vibration Sensor (M-12)     | 09:45 AM<br>01 May 2025 | Active    |
| ALT-2025-049 | Info     | Scheduled maintenance due for Granulator G-01 | Maintenance System          | 08:30 AM<br>01 May 2025 | Scheduled |
| ALT-2025-048 | Warning  | Quality score dropped below target in Line 3  | Quality Analysis Engine     | 07:50 AM<br>01 May 2025 | Active    |
| ALT-2025-047 | Info     | New batch BATCH-2025-003 started in Line 1    | Production System           | 07:30 AM<br>01 May 2025 | Cleared   |
| ALT-2025-046 | Resolved | Humidity level back to normal in Warehouse    | Humidity Sensor (Warehouse) | 06:40 AM<br>01 May 2025 | Resolved  |

Showing 1 to 6 of 18 entries

6 per page

<

1

2

3

>

Alert Details

Critical Alert

Active

Alert ID

: ALT-2025-051

Type

: Critical

Source

: Temperature Sensor (Line 2)

Message

: Line 2 temperature exceeded threshold (85°C)

Triggered On

: 01 May 2025, 10:15 AM

Severity

: High

Status

: Active

Recommended Actions

- Check Line 2 temperature control system.
- Verify cooling system functionality.
- Inspect equipment for overheating issues.
- Take immediate corrective action.

✓ Acknowledge Alert

**TC15 – Alert for Abnormal Condition**

| <b>Field</b>    | <b>Details</b>                                                                      |
|-----------------|-------------------------------------------------------------------------------------|
| Test Case ID    | TC15                                                                                |
| Test Scenario   | Generate Quality Alert                                                              |
| Test Steps      | 1. Provide data representing an abnormal production condition. 2. Process the data. |
| Test Data       | Abnormal condition input                                                            |
| Expected Result | System should generate and display an appropriate alert.                            |
| Actual Result   | To be filled after testing                                                          |
| Status          | To be filled as Pass/Fail                                                           |

**TC16 – No Alert for Normal Condition**

| <b>Field</b>    | <b>Details</b>                                                            |
|-----------------|---------------------------------------------------------------------------|
| Test Case ID    | TC16                                                                      |
| Test Scenario   | Verify Normal Condition Handling                                          |
| Test Steps      | 1. Provide valid normal production data. 2. Process the data.             |
| Test Data       | Normal condition input                                                    |
| Expected Result | System should process the data without generating a false abnormal alert. |
| Actual Result   | To be filled after testing                                                |
| Status          | To be filled as Pass/Fail                                                 |

**5.7 Batch Traceability Test Cases**

**Smart Pharma QA**  
Quality Assurance Platform

Dashboard

Production Monitoring

AI Analysis

Alerts 3

**Batch Traceability**

Reports

Users

Settings

Logout

**Batch Traceability**  
Track and trace batch information across the production lifecycle.

Filters

+ Add New Batch

| Batch ID        | Product Name        | Product Code | Manufacture Date | Expiry Date | Status        | Quality Score | Actions |
|-----------------|---------------------|--------------|------------------|-------------|---------------|---------------|---------|
| BATCH-2025-0001 | Paracetamol 500mg   | PRC-500      | 01 May 2025      | 30 Apr 2027 | Released      | 95%           |         |
| BATCH-2025-0002 | Amoxicillin 250mg   | AMX-250      | 30 Apr 2025      | 29 Apr 2027 | Released      | 92%           |         |
| BATCH-2025-0003 | Ibuprofen 200mg     | IBU-200      | 28 Apr 2025      | 27 Apr 2027 | In Production | 88%           |         |
| BATCH-2025-0004 | Ciprofloxacin 500mg | CIP-500      | 27 Apr 2025      | 26 Apr 2027 | On Hold       | 75%           |         |
| BATCH-2025-0005 | Metformin 500mg     | MET-500      | 25 Apr 2025      | 24 Apr 2027 | Released      | 93%           |         |

Showing 1 to 5 of 25 entries

5 per page

1
2
3
...
5

**Batch Details** (BATCH-2025-0003)

Product Name

Product Code

Batch ID

Manufacture Date

Expiry Date

Manufacturing Line

Packaging Type

Batch Quantity

Current Status

Quality Score

Ibuprofen 200mg

IBU-200

BATCH-2025-0003

28 Apr 2025, 08:30 AM

27 Apr 2027

Line 2

Blister Pack

50,000 Units

In Production

88%

**Process Timeline**

Raw Material Received

Dispensing Completed

Compression Completed

Coating In Progress

Packaging Pending

Quality Check Pending

Release Pending

26 Apr 2025 08:00 AM

26 Apr 2025 10:15 AM

27 Apr 2025 02:30 PM

28 Apr 2025 09:10 AM

-

-

-

Completed

In Progress

Pending

**Related Documents**

Batch Manufacturing Record

Test Report

Certificate of Analysis

Packaging Record

BMR\_B2025-0003.pdf

TR\_B2025-0003.pdf

COA\_B2025-0003.pdf

PACK\_B2025-0003.pdf

Download

Download

Download

Download

## TC17 – Store Batch Information

| Field           | Details                                                      |
|-----------------|--------------------------------------------------------------|
| Test Case ID    | TC17                                                         |
| Test Scenario   | Store Valid Batch Record                                     |
| Test Steps      | 1. Enter valid batch details. 2. Save the batch information. |
| Test Data       | Valid batch ID and associated information                    |
| Expected Result | System should successfully store the batch record.           |
| Actual Result   | To be filled after testing                                   |
| Status          | To be filled as Pass/Fail                                    |

## TC18 – Retrieve Existing Batch

| Field           | Details                                                           |
|-----------------|-------------------------------------------------------------------|
| Test Case ID    | TC18                                                              |
| Test Scenario   | Retrieve Batch Information                                        |
| Test Steps      | 1. Enter an existing batch ID. 2. Search for the batch.           |
| Test Data       | Valid existing batch ID                                           |
| Expected Result | System should retrieve and display the correct batch information. |
| Actual Result   | To be filled after testing                                        |
| Status          | To be filled as Pass/Fail                                         |

## TC19 – Search Non-Existing Batch

| Field        | Details |
|--------------|---------|
| Test Case ID | TC19    |

| Field           | Details                                                                                  |
|-----------------|------------------------------------------------------------------------------------------|
| Test Scenario   | Invalid Batch Search                                                                     |
| Test Steps      | 1. Enter a batch ID that does not exist. 2. Click Search.                                |
| Test Data       | Non-existing batch ID                                                                    |
| Expected Result | System should display a suitable message such as “Batch not found” and should not crash. |
| Actual Result   | To be filled after testing                                                               |
| Status          | To be filled as Pass/Fail                                                                |

## 5.8 Report Generation Test Cases


**Recent Reports**
[View All Reports](#)

| Report Name             | Report Type | Generated By     | Generated On          | Period                    | Actions                                       |
|-------------------------|-------------|------------------|-----------------------|---------------------------|-----------------------------------------------|
| Quality Summary Report  | Summary     | Quality Engineer | 01 May 2025, 10:30 AM | 01 Apr 2025 - 01 May 2025 | <a href="#">Download</a> <a href="#">View</a> |
| Defect Analysis Report  | Detailed    | Quality Engineer | 01 May 2025, 08:15 AM | 01 Apr 2025 - 01 May 2025 | <a href="#">Download</a> <a href="#">View</a> |
| Batch Production Report | Batch       | System           | 30 Apr 2025, 08:00 PM | 01 Apr 2025 - 30 Apr 2025 | <a href="#">Download</a> <a href="#">View</a> |
| AI Analysis Report      | AI / ML     | AI Engine        | 30 Apr 2025, 07:30 PM | 01 Apr 2025 - 30 Apr 2025 | <a href="#">Download</a> <a href="#">View</a> |
| Compliance Report       | Compliance  | Quality Manager  | 30 Apr 2025, 05:45 PM | 01 Apr 2025 - 30 Apr 2025 | <a href="#">Download</a> <a href="#">View</a> |

### TC20 – Generate Quality Report

| Field           | Details                                                                         |
|-----------------|---------------------------------------------------------------------------------|
| Test Case ID    | TC20                                                                            |
| Test Scenario   | Generate Report for Valid Data                                                  |
| Test Steps      | 1. Select a valid batch or data range. 2. Request report generation.            |
| Test Data       | Valid available batch/quality information                                       |
| Expected Result | System should generate the requested quality or compliance report successfully. |
| Actual Result   | To be filled after testing                                                      |
| Status          | To be filled as Pass/Fail                                                       |

### TC21 – Generate Report with No Data

| Field           | Details                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------|
| Test Case ID    | TC21                                                                                        |
| Test Scenario   | Report Generation Without Available Data                                                    |
| Test Steps      | 1. Select a batch or period with no available data. 2. Request report generation.           |
| Test Data       | No matching records                                                                         |
| Expected Result | System should display an appropriate message and should not generate incorrect information. |
| Actual Result   | To be filled after testing                                                                  |
| Status          | To be filled as Pass/Fail                                                                   |

## 5.9 Logout Test Case

### TC22 – Valid Logout

| Field           | Details                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------|
| Test Case ID    | TC22                                                                                                              |
| Test Scenario   | User Logout                                                                                                       |
| Test Steps      | 1. Log in successfully. 2. Click Logout.                                                                          |
| Test Data       | Authenticated user                                                                                                |
| Expected Result | User session should end successfully and protected pages should no longer be accessible without logging in again. |
| Actual Result   | To be filled after testing                                                                                        |
| Status          | To be filled as Pass/Fail                                                                                         |

## 6. NORMAL, BOUNDARY, INVALID AND EXCEPTIONAL CONDITIONS

The assignment requires coverage of normal, boundary, invalid, and exceptional conditions.

### 6.1 Normal Conditions

| Test Case | Condition                    |
|-----------|------------------------------|
| TC01      | Valid user login             |
| TC05      | Dashboard loads successfully |
| TC07      | Valid production data        |
| TC10      | Valid image analysis         |
| TC13      | Valid quality prediction     |
| TC17      | Valid batch storage          |
| TC18      | Valid batch retrieval        |
| TC20      | Valid report generation      |
| TC22      | Valid logout                 |

## 6.2 Boundary Conditions

Boundary testing should be performed using the **actual configured limits of your implemented system**.

Examples:

| Test ID | Boundary Condition               | Expected Result                                            |
|---------|----------------------------------|------------------------------------------------------------|
| BVA01   | Minimum accepted parameter value | System should accept if within valid limit                 |
| BVA02   | Maximum accepted parameter value | System should accept if within valid limit                 |
| BVA03   | Value just below minimum         | System should reject or flag according to validation rules |
| BVA04   | Value just above maximum         | System should reject or flag according to validation rules |
| BVA05   | Minimum accepted batch ID length | System should process according to validation rules        |
| BVA06   | Maximum accepted batch ID length | System should process according to validation rules        |

**Note:** Replace “minimum” and “maximum” with the real limits configured in your project.

## 6.3 Invalid Conditions

| Test Case | Invalid Condition        |
|-----------|--------------------------|
| TC02      | Incorrect password       |
| TC03      | Invalid username         |
| TC04      | Empty login fields       |
| TC11      | Unsupported file format  |
| TC14      | Invalid prediction input |
| TC19      | Non-existing batch ID    |

## 6.4 Exceptional Conditions

| Test Case | Exceptional Condition                |
|-----------|--------------------------------------|
| TC08      | Missing production data              |
| TC12      | Corrupted image file                 |
| TC21      | Report requested when no data exists |

Additional exceptional conditions can be tested in the implemented system, such as unavailable services or failed database requests, if those conditions are supported by your actual project.

## 7. TEST DESIGN TECHNIQUES

The assignment specifically mentions Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition Testing as appropriate techniques.

### 7.1 EQUIVALENCE PARTITIONING

Equivalence Partitioning divides input values into groups that are expected to behave similarly.

Example: Login Testing

| Partition           | Input                             | Expected Result    |
|---------------------|-----------------------------------|--------------------|
| Valid Partition     | Valid username + valid password   | Login successful   |
| Invalid Partition 1 | Valid username + invalid password | Login denied       |
| Invalid Partition 2 | Invalid username + valid password | Login denied       |
| Invalid Partition 3 | Empty username/password           | Validation message |

Related Test Cases

**TC01, TC02, TC03 and TC04** apply Equivalence Partitioning.

### 7.2 BOUNDARY VALUE ANALYSIS

Boundary Value Analysis tests values at the limits of an acceptable input range.

For a production parameter with configured valid range:

Minimum Limit  $\leq$  Parameter Value  $\leq$  Maximum Limit

The following values should be tested:

1. Minimum limit
2. Minimum limit + 1 unit
3. Maximum limit - 1 unit
4. Maximum limit
5. Just below minimum limit

## 6. Just above maximum limit

| Test Value         | Expected Result                          |
|--------------------|------------------------------------------|
| Below minimum      | Reject or flag according to system rules |
| Minimum            | Accept                                   |
| Just above minimum | Accept                                   |
| Just below maximum | Accept                                   |
| Maximum            | Accept                                   |
| Above maximum      | Reject or flag according to system rules |

Related Test Cases

**BVA01 to BVA06.**

## 7.3 DECISION TABLE TESTING

Decision Table Testing is useful for conditions where multiple inputs determine the system output.

Example: Alert Generation

| Rule | Parameter Normal | AI Defect Detected | Expected Alert                                                     |
|------|------------------|--------------------|--------------------------------------------------------------------|
| R1   | Yes              | No                 | No abnormal alert                                                  |
| R2   | No               | No                 | Generate abnormal parameter alert                                  |
| R3   | Yes              | Yes                | Generate defect alert                                              |
| R4   | No               | Yes                | Generate critical/combined quality alert according to system rules |

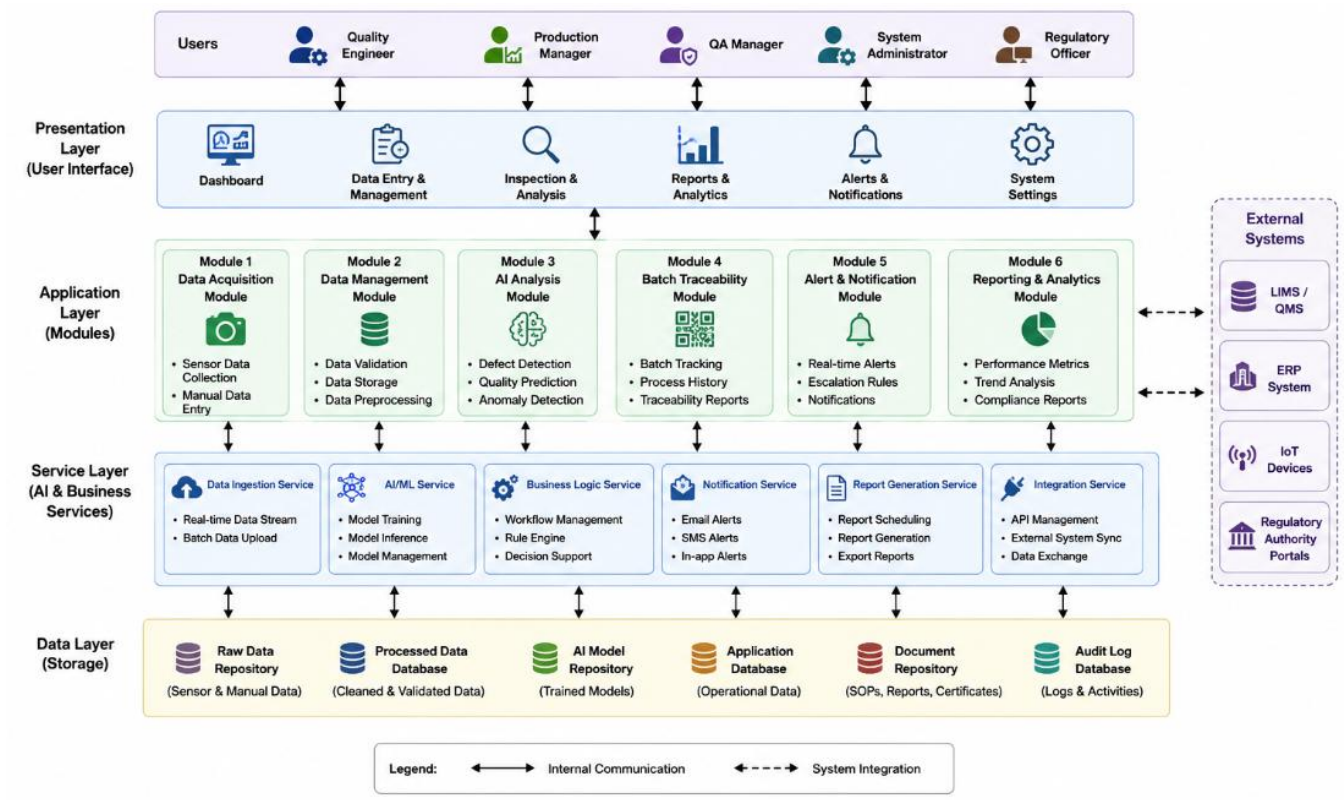
This table ensures that important combinations of production status and defect status are tested.

Related Test Cases

**TC09, TC15 and TC16.**

## 7.4 STATE TRANSITION TESTING

State Transition Testing verifies how the system changes from one state to another.



State Transition Table

| Current State | Event                        | Next State               |
|---------------|------------------------------|--------------------------|
| Logged Out    | Valid credentials            | Logged In                |
| Logged Out    | Invalid credentials          | Login Error              |
| Login Error   | Retry with valid credentials | Logged In                |
| Logged In     | Logout                       | Logged Out               |
| Logged Out    | Attempt protected access     | Access Denied/Login Page |

Related Test Cases

TC01, TC02, TC03, TC04, TC06 and TC22.

## 8. REQUIREMENT TRACEABILITY MATRIX

A Requirement Traceability Matrix maps requirements to test scenarios and test cases. This supports the assignment requirement for adequate requirement and functional coverage.

| Requirement ID | Requirement                  | Test Scenario | Test Cases                               |
|----------------|------------------------------|---------------|------------------------------------------|
| FR01           | User login                   | TS01          | TC01, TC04                               |
| FR02           | Credential validation        | TS02          | TC02, TC03, TC04                         |
| FR03           | Dashboard access             | TS03          | TC05, TC06                               |
| FR04           | Production monitoring        | TS04          | TC07, TC08                               |
| FR05           | Abnormal condition detection | TS05          | TC09                                     |
| FR06           | AI defect detection          | TS06          | TC10, TC11, TC12                         |
| FR07           | Quality prediction           | TS07          | TC13, TC14                               |
| FR08           | Alert generation             | TS08          | TC15, TC16                               |
| FR09           | Batch storage                | TS09          | TC17                                     |
| FR10           | Batch traceability           | TS10          | TC18, TC19                               |
| FR11           | Report generation            | TS11          | TC20, TC21                               |
| FR12           | User logout                  | TS12          | TC22                                     |
| NFR03          | Invalid input handling       | TS13, TS14    | TC04, TC08, TC11, TC14                   |
| NFR04          | Security/access control      | TS16          | TC06                                     |
| NFR05          | Reliability                  | TS15          | TC05, TC07, TC17                         |
| NFR06          | Error messages               | TS14, TS17    | TC02, TC03, TC04, TC11, TC12, TC19, TC21 |
| NFR10          | Exception handling           | TS17          | TC08, TC12, TC21                         |

## 9. TEST COVERAGE SUMMARY

| Category                 | Test Cases Covered        |
|--------------------------|---------------------------|
| Authentication           | TC01–TC04, TC22           |
| Dashboard                | TC05–TC06                 |
| Production Monitoring    | TC07–TC09                 |
| AI Defect Detection      | TC10–TC12                 |
| Quality Prediction       | TC13–TC14                 |
| Alert Generation         | TC15–TC16                 |
| Batch Management         | TC17–TC19                 |
| Report Generation        | TC20–TC21                 |
| Normal Conditions        | Covered                   |
| Boundary Conditions      | Covered through BVA cases |
| Invalid Conditions       | Covered                   |
| Exceptional Conditions   | Covered                   |
| Equivalence Partitioning | Applied                   |
| Boundary Value Analysis  | Applied                   |

| Category                 | Test Cases Covered |
|--------------------------|--------------------|
| Decision Table Testing   | Applied            |
| State Transition Testing | Applied            |

## 10. TEST EXECUTION RESULT FORMAT

After you actually test the project, you can fill the following summary.

| Test Case ID | Expected Result                  | Actual Result | Status    |
|--------------|----------------------------------|---------------|-----------|
| TC01         | User logs in successfully        |               | Pass/Fail |
| TC02         | Invalid password rejected        |               | Pass/Fail |
| TC03         | Invalid username rejected        |               | Pass/Fail |
| TC04         | Empty fields validated           |               | Pass/Fail |
| TC05         | Dashboard loads                  |               | Pass/Fail |
| TC06         | Unauthorized access denied       |               | Pass/Fail |
| TC07         | Valid production data displayed  |               | Pass/Fail |
| TC08         | Missing data handled             |               | Pass/Fail |
| TC09         | Abnormal condition detected      |               | Pass/Fail |
| TC10         | Valid image analyzed             |               | Pass/Fail |
| TC11         | Unsupported file rejected        |               | Pass/Fail |
| TC12         | Corrupted file handled           |               | Pass/Fail |
| TC13         | Quality prediction generated     |               | Pass/Fail |
| TC14         | Invalid prediction data rejected |               | Pass/Fail |
| TC15         | Alert generated                  |               | Pass/Fail |
| TC16         | No false alert                   |               | Pass/Fail |
| TC17         | Batch stored                     |               | Pass/Fail |
| TC18         | Batch retrieved                  |               | Pass/Fail |
| TC19         | Missing batch handled            |               | Pass/Fail |
| TC20         | Report generated                 |               | Pass/Fail |
| TC21         | No-data report request handled   |               | Pass/Fail |
| TC22         | User logged out                  |               | Pass/Fail |

## 11. CONCLUSION



The AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform successfully integrates artificial intelligence and real-time analytics into pharmaceutical production processes to ensure superior product quality, regulatory compliance, and operational efficiency. By leveraging automated defect detection, batch traceability, and intelligent alerts, the system enhances decision-making and minimizes human error, ultimately leading to safer medicines and improved patient outcomes.

### Key Outcomes Achieved



**Improved Quality Assurance:** AI-powered defect detection and predictive analytics significantly improve the accuracy and consistency of quality inspections.



**Operational Efficiency:** Automation of data collection, analysis, and reporting reduces manual effort, shortens inspection time, and increases overall productivity.



**Enhanced Traceability:** End-to-end batch traceability ensures complete visibility of production processes, facilitating quick root cause analysis and regulatory compliance.



**Proactive Risk Management:** Real-time alerts and anomaly detection enable early identification of issues, preventing potential quality failures and reducing product recalls.



**Data-Driven Decision Making:** Comprehensive analytics dashboards and reports provide valuable insights for continuous improvement and strategic decision-making.



#### Overall Impact

The proposed platform bridges the gap between traditional quality control and modern intelligent systems. It ensures consistent product quality, regulatory adherence, and operational excellence, making pharmaceutical manufacturing smarter, safer, and more reliable.



*"Smart Quality Today, Safer Medicines Tomorrow."*

This test case design document identifies the major functional and non-functional requirements of the AI-Based Smart Pharmaceutical Manufacturing Quality Assurance Platform. Comprehensive test scenarios and 22 detailed test cases have been prepared to cover normal, invalid, boundary, and exceptional conditions.

The test design applies multiple techniques, including Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing. A Requirement Traceability Matrix is also included to map system requirements to corresponding test scenarios and test cases.

The final execution results and Pass/Fail status should be completed based on the actual testing of the implemented project. This approach provides structured coverage of the system's functionality, reliability, validation, and error-handling behavior.

The document structure is designed to match the assignment's requirement for comprehensive requirements, scenarios, test data, expected results, test-design techniques, and traceability coverage. The rubric particularly rewards comprehensive normal/boundary/invalid/exceptional coverage, multiple relevant test techniques, strong requirement-to-test mapping, and clear professional presentation.